

MOROGORO, Tanzania, United Republic ofWMO: **638660**Lat: **6.833S**Lon: **37.650E**Elev: **1726**StdP: **13.80**Time Zone: **3.00 (E03)**Period: **06-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	59.2	60.9	53.2	64.3	74.6	54.8	68.0	74.6	13.8	76.6	11.9	76.3	0.0	240	0.240

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.0	93.5	73.5	92.4	73.3	91.3	73.1	77.4	86.8	76.6	86.0	75.9	85.1	4.2	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.1	140.4	81.4	74.0	135.2	79.9	73.6	133.2	79.4	42.2	87.1	41.4	86.1	40.8	85.3	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.3	13.9	11.8	DB	55.5	97.0	0.6	1.8	55.1	98.3	54.7	99.3	54.3	100.4	53.9	101.7
			WB	53.6	81.5	2.0	2.5	52.2	83.3	51.1	84.8	50.0	86.1	48.5	87.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	77.2	81.3	81.0	80.0	77.4	75.3	72.6	71.9	73.4	75.3	77.8	79.7	80.9
	DBStd	3.87	2.41	2.05	2.27	1.86	1.71	2.10	1.94	1.78	1.89	1.83	2.18	2.16
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9931	969	869	931	822	786	679	679	725	759	863	892	957
	CDD65	4456	504	449	466	372	321	229	214	260	309	398	442	492
	CDH74	36045	4593	4217	3913	2178	1681	1403	1410	1877	2622	3513	4079	4559
	CDH80	13032	1974	1851	1583	574	310	213	196	373	841	1412	1775	1930
(14) Wind	WSAvg	2.2	4.6	3.6	1.5	0.3	0.2	0.3	0.8	1.6	3.0	3.0	3.9	4.1
(15) Precipitation	PrecAvg	30.4	3.8	3.5	4.4	7.1	2.5	0.6	0.5	0.2	0.4	1.1	2.3	4.3
	PrecMax	46.9	8.6	8.0	9.4	12.5	5.9	2.0	2.6	1.3	3.0	4.0	7.9	11.8
	PrecMin	17.0	0.4	0.0	0.9	3.5	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.6
	PrecStd	7.6	2.6	2.2	2.1	2.3	1.4	0.6	0.7	0.4	0.7	1.1	2.2	3.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.0	95.1	94.2	89.4	86.2	85.8	85.3	86.4	89.8	92.2	93.6	94.0
		MCWB	72.8	74.1	75.6	75.4	72.5	69.7	68.1	68.4	70.0	71.5	71.9	73.9
	2%	DB	93.2	93.2	91.8	87.3	84.5	84.0	83.8	84.6	88.1	90.3	91.8	92.4
		MCWB	72.9	73.9	74.6	75.0	72.5	69.4	67.5	67.8	69.4	71.0	72.1	73.5
	5%	DB	91.5	91.4	90.0	85.6	83.2	82.5	82.3	83.9	86.4	89.2	91.0	91.1
		MCWB	73.2	73.6	74.1	74.9	72.3	69.0	67.3	67.5	68.7	70.8	72.1	73.3
	10%	DB	89.6	89.6	88.0	83.9	82.0	80.9	80.7	82.3	84.6	87.4	89.2	89.3
		MCWB	73.4	73.3	74.3	74.4	72.1	68.6	67.1	67.1	68.1	70.3	72.1	73.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.4	77.6	77.9	77.5	75.6	72.9	72.2	71.3	73.5	76.7	78.0	78.4
		MCDB	87.7	88.1	88.7	85.0	81.6	80.6	80.9	80.8	85.3	86.4	86.8	87.4
	2%	WB	76.3	76.4	76.9	76.5	74.6	72.0	70.4	70.1	71.3	73.7	76.1	76.8
		MCDB	87.0	87.5	87.1	83.8	80.5	80.0	79.2	79.8	82.9	83.5	85.2	86.0
	5%	WB	75.5	75.5	76.2	75.8	73.8	71.0	69.2	69.1	70.2	72.6	74.7	75.5
		MCDB	86.0	86.6	85.6	82.6	79.8	78.9	78.0	78.9	81.6	82.6	84.1	85.3
	10%	WB	74.7	74.6	75.4	75.1	73.1	70.2	68.3	68.3	69.3	71.7	73.6	74.6
		MCDB	84.6	85.1	84.0	81.5	78.9	77.9	77.2	78.1	80.4	82.0	83.0	84.3
(35) Mean Daily Temperature Range	5% DB	MDBR	17.6	18.0	17.1	14.1	14.5	18.8	19.7	19.5	21.3	20.4	19.3	17.9
		MCDBR	19.5	20.6	19.8	16.5	16.7	20.2	21.6	21.3	23.1	22.9	21.9	20.0
	5% WB	MCWBR	5.2	5.7	5.8	6.6	6.9	8.4	8.8	7.1	7.2	6.7	5.6	5.3
		MCDWR	17.4	18.3	17.4	14.4	13.7	17.8	18.8	19.3	21.5	19.9	18.6	17.7
(40) Clear-Sky Solar Irradiance	taub		0.446	0.450	0.409	0.364	0.362	0.368	0.395	0.406	0.399	0.392	0.386	0.412
		taud	2.318	2.310	2.441	2.598	2.557	2.486	2.364	2.332	2.355	2.394	2.453	2.422
	Ebn at Noon		284	282	290	295	287	280	274	278	288	296	300	293
		Edh at Noon	43	44	38	31	31	32	37	40	41	40	38	39
(44) All-Sky Solar Radiation	RadAvg		1842	1882	1759	1529	1439	1436	1426	1469	1643	1750	1828	1853
	RadStd		143	112	115	99	108	66	73	67	98	102	94	103

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page